

# Non-stationary simulation study: what a mean-only MRTS augmentation can and cannot buy

July 2026 — interim report v4 (settings 1–12, the threshold-vs-symmetry disambiguation, the  $K \geq 30$  extension, and the non- $t$  extension 13–21; settings 16 and 20 at  $n = 10$ )

## Abstract

We evaluate the MRTS covariate extension of the Morris et al. (2017) skew- $t$  space-time model on twelve simulated data-generating processes (DGPs) that place non-stationarity in different moments: the mean (settings 1, 2, 11, 12), the spatial dependence (settings 3–8), the tails (setting 9), and the skewness (setting 10). Three findings. (i) MRTS improves predictive scores *only* where the truth carries time-averaged mean structure, and this is provable in advance: an oracle calculation shows the relative Brier score has a floor  $R^* \approx 1$  for nine of eleven settings, so their observed nulls are properties of the design, not evidence about MRTS. (ii) Where mean structure exists, the energy- and variogram-score gains are carried *entirely* by the two thresholded symmetric- $t$  methods. A disambiguating pair of methods (symmetric- $t$  *without* threshold,  $n = 10$ ) attributes this cleanly to the *threshold*, not the symmetry — and the absolute scores reframe it: POT censoring without a mean basis damages the predictive field so badly (baseline energy  $2.6 \times -16 \times$  worse) that MRTS is a partial *rescue*, not a synergy; the unthresholded methods never needed rescuing and remain better in absolute terms at every  $K$ . (iii) The Brier score is close to blind to mean recovery and can even deteriorate with  $K$ : in the strong-mean setting 11 a fit that recovers every parameter of the DGP at  $K = 25$  has a *worse* Brier score than the badly misspecified  $K = 0$  baseline. The rich-mean setting 12, run out to  $K = 50$ , resolves this: the deterioration is confined to intermediate  $K$  (where the mean basis is still too coarse), and once the basis is fine enough ( $K \geq 30$ –40) the Brier score improves by 7–12%, consistent with the extrapolation-error mechanism. The discriminating check passes: the simple-surface setting 11, extended to the same  $K = 50$ , never turns positive — its Brier ratio only decays back to parity ( $1.09 \rightarrow 1.00$ ) — so the reversal requires mean structure that only a fine basis can resolve, not merely a large  $K$ . New in v4: the nine non- $t$  extension settings (13–21; Gaussian-GP errors, so the skew- $t$  error law is misspecified) are fitted and included in the score tables — settings 16 and 20 at  $n = 10$  on the full  $K$  grid, the other seven as  $n = 3$  pilots; their systematic analysis is deferred.

## 1 Design

**Data-generating process.** Settings 1–12 share the skew- $t$ -1 core on  $n_s = 144$  sites drawn uniformly on  $[0, 10]^2$  (settings 13–21 replace it with a Gaussian process; see below):

$$Y_t(\mathbf{s}) = 10 + \lambda \sigma_t |z_t| + \sigma_t v_t(\mathbf{s}) + [\text{structure}]_k, \quad t = 1, \dots, 50, \quad (1)$$

with  $\lambda = 3$ ,  $z_t \stackrel{iid}{\sim} N(0, 1)$  half-normal,  $v_t \sim N(0, C)$  exponential correlation ( $\rho = 1$ ,  $\gamma = 0.9$ ),  $\tau_t \sim \text{Ga}(3/2, 4)$ ,  $\sigma_t = \tau_t^{-1/2}$  (marginal  $t_3$ ), i.i.d. in time. The settings differ only in the structure term—or, for the *replace- $C$*  settings, in the error correlation itself (Table 1). Table 2 gives the

resulting anatomy of each DGP: the marginal law of  $Y_t(\mathbf{s})$ , the spatial dependence of the error process, and the time-averaged true mean surface  $\bar{\mu}(\mathbf{s})$  (the target of any pooled- $\beta$  mean basis). Settings 13–21 form a *non- $t$  extension*: they drop the skew- $t$  generator entirely in favour of a nonlinear mean surface (sd 11) with Gaussian-GP errors ( $\sigma_g = 4$ , same  $C_{\text{exp}}$ ), so that the mean channel is tested without the crutch of a correctly specified error law at large  $K$ . All nine are fitted and scored and appear in the score tables — settings 16 and 20 at  $n = 10$  on the full  $K$  grid, the other seven as  $n = 3$  pilots ( $K_{\text{MRTS}} \in \{0, 30, 40, 50\}$ ); Section 5 introduces four of them in detail, and the systematic analysis is deferred to the next revision.

Table 1: The DGP settings and the number of simulation replicates  $n$  (independent datasets) fitted per setting. Settings 1–12 share the skew- $t$  core; settings 13–21 are the non- $t$  extension (Gaussian-GP errors,  $\sigma_g = 4$ ; nonlinear mean surface orthogonalised against  $[1, s_1, s_2]$  and standardised to sd = 11). Routes: *mean* = added to the fixed mean; *additive* = mean-zero random field added to the draw; *replace  $C$*  = non-stationary correlation handed to the generator (unit diagonal, so the *marginal* skew- $t$  law is unchanged — verified by KS tests,  $p \in [0.075, 0.864]$ ); *skew term* = spatially varying  $\lambda$ .

| ID                                                                                                          | surf_type        | non-stationarity                                            | route       | $n$             |
|-------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------|-------------|-----------------|
| 1                                                                                                           | invariant        | mean, 2 cosine bumps, sd = 3.7                              | mean        | 10              |
| 2                                                                                                           | varying          | mean, time-varying weights                                  | mean        | 10 <sup>a</sup> |
| 3                                                                                                           | ns_dependence    | dependence, low-rank cosine RE                              | additive    | 10              |
| 4                                                                                                           | ps_cov           | dependence, Paciorek–Schervish                              | replace $C$ | 3               |
| 5                                                                                                           | ps_add           | dependence, same PS field                                   | additive    | 3               |
| 6                                                                                                           | covdep_cov       | dependence, $\rho(\mathbf{s}) = e^{0.7f_1(\mathbf{s})}$     | replace $C$ | 3               |
| 7                                                                                                           | covdep_add       | dependence, same field                                      | additive    | 3               |
| 8                                                                                                           | fuentes_cov      | dependence, 4-regime mixture                                | replace $C$ | 3               |
| 9                                                                                                           | gh_marginal      | tails, Tukey $g$ -and- $h$ , $g(\mathbf{s}), h(\mathbf{s})$ | additive    | 3               |
| 10                                                                                                          | lambda_varying   | skewness, $\lambda(\mathbf{s}) = 3 + 3f_2(\mathbf{s})$      | skew term   | 3               |
| 11                                                                                                          | invariant_strong | mean, 3 $\times$ setting 1, sd = 11                         | mean        | 3               |
| 12                                                                                                          | mrts_multibump   | mean, 6 Gaussian bumps, sd = 11                             | mean        | 3               |
| <i>Non-<math>t</math> extension (Gaussian-GP errors; skew-<math>t</math> generator dropped)<sup>b</sup></i> |                  |                                                             |             |                 |
| 13                                                                                                          | franke           | mean, Franke (multiscale)                                   | mean        | 3               |
| 14                                                                                                          | ridge_curved     | mean, curved Gaussian ridge                                 | mean        | 3               |
| 15                                                                                                          | annulus          | mean, Ricker ring ( $r_0=3, w=1.5$ )                        | mean        | 3               |
| 16                                                                                                          | sigmoid_front    | mean, tanh arc front                                        | mean        | 3               |
| 17                                                                                                          | gp_mean          | mean, GP draw (per dataset, fixed in $t$ )                  | mean        | 3               |
| 18                                                                                                          | franke_sas       | mean = setting 13; sinh-arcsinh error                       | mean        | 3               |
| 19                                                                                                          | poly2            | mean, bowl + saddle (2nd-order poly)                        | mean        | 3               |
| 20                                                                                                          | transform_mix    | mean, log + exp-decay + ratio terms                         | mean        | 3               |
| 21                                                                                                          | nonsmooth        | mean, hinge + $ \cdot $ ( $C^0$ creases)                    | mean        | 3               |

<sup>a</sup> setting 2 was fitted at  $K = 0$  only (no MRTS sweep); it is excluded from the score tables. <sup>b</sup> settings 13–21: settings 16 and 20 at  $n = 10$  on the full  $K$  grid, the other seven as  $n = 3$  pilots on  $K_{\text{MRTS}} \in \{0, 30, 40, 50\}$ .

**Fitted methods and grid.** Five baseline methods are fitted to every dataset: (1) Gaussian; (2) skew- $t$ ,  $K_{\text{knots}} = 1$ ; (3) symmetric- $t$ ,  $K_{\text{knots}} = 1$ , POT threshold at  $q(0.80)$ ; (4) skew- $t$ ,  $K_{\text{knots}} = 5$ ; (5) symmetric- $t$ ,  $K_{\text{knots}} = 5$ , threshold  $q(0.80)$ . Each is run across MRTS design widths  $K_{\text{MRTS}} \in \{0, 3, 5, 8, 10, 12, 15, 20, 25\}$ , where  $K_{\text{MRTS}} = 0$  denotes the baseline design  $[1, s_1, s_2]$  and  $K_{\text{MRTS}} = K$  replaces the design with the  $K$ -column `autoFRK::mrts` basis built on the *training* sites only (column 1 = intercept, 2–3 linear, 4: $K$  thin-plate-spline eigenfunctions;  $K = 3$  spans the baseline).

MCMC: 20,000 iterations, 10,000 burn-in. The random seed is shared across  $K_{\text{MRTS}}$  within each (method, dataset), so all  $K$ -comparisons are *paired*. The  $K_{\text{MRTS}} \in \{30, 40, 50\}$  extension is now complete for every fitted skew- $t$  setting (1, 3–12), so the full 12-point grid  $\{0, 3, \dots, 50\}$  underlies all their tables.

**Scoring.** The last 44 of the 144 sites are held out entirely; all scores are computed out-of-sample on those sites. With  $\hat{F}$  the posterior predictive law:

$$\text{BS} = \frac{1}{11} \sum_p \frac{1}{44 \cdot 50} \sum_{i,t} \left( \hat{P}(Y_t(\mathbf{s}_i^*) > u_p) - \mathbb{1}\{y_t(\mathbf{s}_i^*) > u_p\} \right)^2, \quad p \in \{0.90, \dots, 0.995\}, \quad (2)$$

$$\text{ES} = \frac{1}{50} \sum_t \left[ \mathbb{E} \|\mathbf{X}_t - \mathbf{y}_t\| - \frac{1}{2} \mathbb{E} \|\mathbf{X}_t - \mathbf{X}'_t\| \right], \quad \mathbf{X}_t, \mathbf{X}'_t \sim \hat{F}_t \text{ (44-dim)}, \quad (3)$$

$$\text{VS} = \frac{1}{50} \sum_t \sum_{i < j} \left( |y_{it} - y_{jt}|^{1/2} - \mathbb{E} |X_{it} - X_{jt}|^{1/2} \right)^2. \quad (4)$$

The Brier score is a per-cell *marginal* score and cannot see spatial dependence; the energy score is the multivariate CRPS (reduces to CRPS at  $d = 1$ ) and is location-sensitive; the variogram score ( $p = 1/2$ ) targets pairwise dependence. Every result below is the *relative* score  $R_{m,K,d} = S_{m,K,d}/S_{m,0,d}$  against the same method’s own  $K = 0$  fit, averaged over datasets;  $R < 1$  means MRTS helps.

## 2 Results

Tables 3–5 report, per setting and method, the best (smallest)  $\bar{R}_{m,K}$  over  $K > 0$ , with the minimising  $K$  in parentheses. Asterisks mark  $|\bar{R} - 1| > 2\widehat{\text{SE}}$  (SE over datasets; with  $n = 3$  this is indicative only). Table 6 gives the full setting  $\times K$  profile of the relative Brier score, method by method.

### 2.1 Finding 1: gains appear only where the truth has mean structure — and this is provable a priori

The only large entries in any table are settings 1, 11 and 12 (mean structure), methods 3 and 5: energy and variogram reductions of 34–42% (setting 1), 73–80% (setting 11) and 58% (setting 12). Every dependence, tails and skewness setting is flat to within a few percent, in *all three* scores.

This dichotomy can be established without MCMC. Because the MRTS coefficient  $\beta$  is pooled over time, the most any mean basis can deliver is the time-averaged true mean  $\bar{\mu}(\mathbf{s})$ . Define the oracle relative Brier

$$R^* = \frac{\text{BS}(\bar{\mu})}{\text{BS}(\text{spatial constant})}, \quad (5)$$

both evaluated under the *true* skew- $t$  predictive law.  $R^*$  is a floor: no  $K$ , sample size or sampler can push the relative Brier below it. Table 9 shows  $R^* \approx 1$  for nine of eleven settings: their Brier nulls are design properties, not evidence about MRTS. Setting 10 is the striking case:  $R^* = 1.12 > 1$  — supplying the *true* mean makes the Brier score *worse*, because the mean shift induced by  $\lambda(\mathbf{s})$  is a symptom of a skewness change that the constant- $\lambda$  model cannot represent; correcting the symptom while the disease remains actively misleads the tail probabilities.

## 2.2 Finding 2: the gains are carried entirely by the thresholded methods

In Table 3, setting 11 (mean sd = 11), the two POT-thresholded symmetric- $t$  methods reduce the energy score by 73–80%, while the Gaussian and both skew- $t$  methods stay within 2% of their  $K = 0$  baselines — despite those same skew- $t$  fits recovering the mean surface almost exactly (recovery RMSE 14.0  $\rightarrow$  0.66 for method 2 at  $K = 25$ ) and recovering every DGP parameter (Section 2.3). Table 10 shows the full  $K$  profile.

Methods 3/5 are simultaneously (a) POT-thresholded and (b) symmetric- $t$ , so a disambiguating pair was added: methods 11/12, *identical to methods 3/5 except* threshold = 0, fitted on setting 1 with  $n = 10$  (Table 11). The verdict is unambiguous: the unthresholded symmetric- $t$  methods are *flat* (energy ratio 0.99 at every  $K$ , indistinguishable from the skew- $t$  reference), while their thresholded twins gain 35–40%. **The threshold, not the symmetry, is the driver.**

The absolute column reframes the finding. The thresholded baselines are *catastrophically* worse to begin with: censoring everything below  $q(0.80)$  leaves the constant-mean fit with almost no information about the spatial level of the field, and its out-of-sample predictive field degrades by a factor 2.6 (K1) to 16 (K5) in energy score. MRTS repairs the mean that the censored likelihood cannot learn — a partial *rescue*, not a synergy: even at the best  $K$ , method 3 reaches  $21.8 \times 0.579 \approx 12.6$ , still well above the 8.5 that methods 2/11 achieve *without any MRTS at all*. In absolute terms the unthresholded methods win at every  $K$ ; the correct reading of Finding 2 is therefore “*only the thresholded methods have room to be helped*,” and the practical implication is for POT-style threshold models specifically: if the mean is not modelled well, censored fitting amplifies the damage, and a mean basis buys back part of it.

## 2.3 Finding 3: the Brier score is nearly blind to mean recovery, and can punish it

Setting 11, method 2 (skew- $t$  K1), dataset 1, posterior means (truth:  $\lambda = 3$ ,  $\mathbb{E}[\sigma] = 2.26$ ,  $\rho = 1$ ,  $\gamma = 0.9$ ):

| $K_{\text{MRTS}}$ | recovery RMSE | $\hat{\lambda}$ | $\mathbb{E}[\hat{\sigma}]$ | $\hat{\rho}$ | $\hat{\gamma}$ |
|-------------------|---------------|-----------------|----------------------------|--------------|----------------|
| 0                 | 14.01         | 0.15            | 11.16                      | 2.04         | 0.99           |
| 10                | 2.51          | 2.04            | 3.01                       | 1.21         | 0.91           |
| 25                | <b>0.66</b>   | <b>3.18</b>     | <b>1.94</b>                | <b>0.96</b>  | <b>0.94</b>    |

At  $K = 0$  the model cannot represent the sd-11 mean surface and absorbs the spatial variance into an inflated  $\sigma$  ( $5\times$  truth) with  $\lambda$  crushed to zero. At  $K = 25$  *every* parameter is recovered. Yet the  $K = 25$  Brier score is *worse* than  $K = 0$  by 8.6% (Table 10), and the same reversal holds for methods 1, 4, 5. A correctly specified model loses to a badly misspecified one on this score. The working hypothesis — not yet verified — is an out-of-sample interaction: the sharpened predictive distribution ( $\hat{\sigma}$  correct instead of inflated) exposes the residual error of the MRTS mean *extrapolated* to the held-out sites, whereas the misspecified  $K = 0$  fit hides that error inside an accidentally conservative (inflated- $\sigma$ ) predictive distribution.

**Setting 12 resolves the anomaly.** The rich-mean setting 12 (six bumps, sd 11) was run out to  $K_{\text{MRTS}} = 50$ , where the noise-free MRTS projection error of its surface finally becomes small (0.71 at  $K = 5$ , 0.13 at  $K = 25$ , 0.057 at  $K = 50$ ). The Brier profile follows exactly the pattern the extrapolation mechanism predicts: flat-to-worse at intermediate  $K$  (ratios 1.00–1.05 for  $K \leq 25$ , where the basis is still too coarse and the residual mean error is exposed by the sharpened predictive), then *improving* once the basis is fine enough:

| relative Brier, setting 12 | $K=15$ | $K=25$ | $K=30$ | $K=50$       | recovery at $K=50$ |
|----------------------------|--------|--------|--------|--------------|--------------------|
| Gaussian                   | 1.019  | 1.045  | 0.988  | <b>0.930</b> | 0.512              |
| Skew- $t$ K1               | 1.022  | 1.020  | 0.980  | <b>0.925</b> | 0.073              |
| Sym- $t$ K1 thr.           | 0.899  | 0.918  | 0.892  | <b>0.880</b> | 0.422              |
| Skew- $t$ K5               | 0.976  | 1.022  | 0.965  | <b>0.913</b> | 0.268              |

(Method 5 is excluded: its exponential-fallback instability blows up one of the three datasets at large  $K$ , ratio 2.7.) The  $K=50$  gains of 7–12% are real but far above the oracle floor  $R^* = 0.495$  — the mean channel is recovered (skew- $t$  K1 recovery ratio 0.073), yet most of the theoretically available Brier headroom is spent compensating the sharpness–extrapolation trade-off. Practically: *for judging mean-structure recovery, the Brier score remains the wrong instrument — it responds only when the basis is fine enough to make the residual extrapolation error negligible; energy/variogram scores (or the recovery RMSE directly) are the informative ones.*

**The large- $K$  limit discriminates the mechanism.** If the reversal were simply a matter of “ $K$  large enough,” setting 11 — whose 2-bump surface the basis captures by  $K \approx 10$  — should also turn positive by  $K = 50$ . It does not (Table 10, rightmost columns): the mid- $K$  deterioration decays monotonically (1.16 at  $K=12$  to 1.01–1.00 at  $K=50$  for the skew- $t$  and Gaussian methods) but never crosses below parity, leaving the oracle headroom  $R^* = 0.439$  entirely unused. The weak-signal setting 1 behaves the same way at large  $K$  (skew- $t$  K1: 0.985 at  $K=25$ , 0.978 at  $K=50$  — smooth, small, no regime change). Same  $K$  grid, same machinery, opposite outcome to setting 12: the Brier reversal requires mean structure that *only* a fine basis can resolve; extra columns on an already-resolved surface buy nothing and merely let the sharpness–extrapolation penalty fade.

### 3 Caveats

- Settings 4–11 are pilots with  $n = 3$ ; settings 1 and 3 have  $n = 10$ . Starred cells with  $n = 3$  should be read as indicative. The small- $n$  *best-over- $K$*  statistic also has a selection bias toward values  $< 1$ ; flat profiles (e.g. Sym- $t$  K5 rows in dependence settings, best  $\approx 0.72$ –0.85 with non-monotone  $K$  profiles) are consistent with noise plus this selection effect.
- Method 5 (Sym- $t$  K5 T) occasionally triggers its exponential-covariance fallback and produces isolated outlier fits (e.g. setting 4,  $K = 25$ : energy ratio 15.8 in one dataset). Best-over- $K$  cells are robust to this; per- $K$  curves at  $n = 3$  are not.
- All  $K$  comparisons are paired (shared seeds across  $K$ ), so within-method ratios are considerably more precise than the between-method comparisons.
- The recovery RMSE (not tabulated here) has a known bimodality on flat-mean settings caused by the intercept- $\lambda$  ridge; see `nonsta_settings_design.md`.

### 4 The setting-12 showcase: why a large $K$ is worth having

Setting 12’s six-bump surface was designed so that the MRTS approximation improves *gradually* with  $K$  (noise-free projection error 0.71 at  $K=5$ , 0.19 at  $K=15$ , 0.057 at  $K=50$ ), in contrast to the 2-bump settings which the basis captures by  $K=5$ . The fitted results track that curve: skew- $t$  K1 recovery ratios are 0.93 ( $K=5$ )  $\rightarrow$  0.49 ( $K=10$ )  $\rightarrow$  0.14 ( $K=25$ )  $\rightarrow$  0.073 ( $K=50$ ) — the extended elbow, realised in MCMC. Two by-products: (a) the Brier anomaly resolution above; (b)

a resolution ceiling — a higher-frequency mean component (3 cycles per axis) is *unrecoverable* on the 100-train/44-test geometry (its projection error *rises* with  $K$ ), i.e. MRTS’s usable resolution is capped by site density, and the six-bump surface sits just inside that cap.

## 5 The non- $t$ spotlight quartet: settings 16, 19, 20, 21

### Shared construction

- All four settings draw

$$Y_t(\mathbf{s}) = 10 + g(\mathbf{s}) + \sigma_g \varepsilon_t(\mathbf{s}), \quad \varepsilon_t \stackrel{iid}{\sim} N(\mathbf{0}, C_{\text{exp}}), \quad \sigma_g = 4, \quad t = 1, \dots, 50, \quad (6)$$

so the marginal law is Gaussian and every  $t$ -based error model is misspecified. Methods, hold-out and scoring are as in Section 1; the fitted grid is  $K_{\text{MRTS}} \in \{0, 30, 40, 50\}$  with  $n = 3$ , except settings 16 and 20, which are at  $n = 10$  on the full  $K$  grid  $\{0, 3, 5, 8, 10, 12, 15, 20, 25, 30, 40, 50\}$ .

- Each mean surface is the image of a raw field  $\tilde{g} \in \mathbb{R}^{144}$  under the orthogonalise-and-standardise map

$$g = 11 \frac{(I - P_X) \tilde{g}}{\widehat{\text{sd}}\{(I - P_X) \tilde{g}\}}, \quad P_X = X(X^\top X)^{-1} X^\top, \quad X = [\mathbf{1}, s_1, s_2] \in \mathbb{R}^{144 \times 3}, \quad (7)$$

so that  $g \perp \text{col}(X)$  and  $\widehat{\text{sd}}(g) = 11$  exactly. Hence the  $K = 0$  design spans *none* of  $g$ ; composite raw terms are standardised to unit sd before summing, written  $\text{std}\{\cdot\}$  below.

- Recoverability is quantified before any MCMC by the noise-free out-of-sample projection error

$$e(K) = \frac{\|B_K^{\text{te}} \hat{\beta}_K - g^{\text{te}}\|_2}{\|g^{\text{te}} - \bar{g}^{\text{te}}\|_2}, \quad \hat{\beta}_K = \arg \min_{\beta \in \mathbb{R}^K} \|B_K^{\text{tr}} \beta - g^{\text{tr}}\|_2, \quad (8)$$

where  $B_K$  is the  $K$ -column MRTS basis built on the 100 training sites (superscripts: evaluated at the training / 44 test sites). Table 12 reports  $e(K)$  together with the oracle Brier floor  $R^*$  recomputed under (6).

### Setting 16 (sigmoid front): the steepest recoverable front

- With  $r(\mathbf{s}) = \|\mathbf{s} - (2, 2)\|_2$  and width  $w = 1$ ,

$$\tilde{g}_{16}(\mathbf{s}) = \tanh(w^{-1}\{r(\mathbf{s}) - 4.5\}). \quad (9)$$

- Since  $\|\nabla \tilde{g}_{16}\| = w^{-1} \text{sech}^2(w^{-1}\{r - 4.5\}) \leq w^{-1}$ , the transition is confined to the annulus  $|r - 4.5| \lesssim w$ , of width  $\approx 2w = 2$  units. The width is calibrated to the site-density resolution cap of Section 5:  $e(K)$  decays monotonically with no extrapolation blow-up (Table 12).
- Role: the sharpest *smooth* localised structure the 100-site training density supports — the fine- $K$  stress test for mean recovery under a misspecified error law.

### Setting 19 (poly2): the easy-end control

- With rescaled coordinates  $(x_1, x_2) = \mathbf{s}/10$ ,

$$\tilde{g}_{19}(\mathbf{s}) = \text{std}\{(x_1 - \frac{1}{2})^2 + (x_2 - \frac{1}{2})^2\} + \text{std}\{(x_1 - \frac{1}{2})(x_2 - \frac{1}{2})\}. \quad (10)$$

- $\tilde{g}_{19}$  lies in the 6-dimensional space  $\mathcal{P}_2$  of quadratic polynomials, and  $(I - P_X)$  quotients out its affine part, so the target is the 3-dimensional pure-quadratic component  $\text{span}\{s_1^2, s_1 s_2, s_2^2\} \bmod \text{col}(X)$ . Accordingly  $e(K)$  collapses fastest of all nine non- $t$  surfaces ( $0.23 \rightarrow 0.05 \rightarrow 0.015$  at  $K = 5, 15, 50$ ).
- Role: control setting — a method that fails here fails for likelihood or sampler reasons, not basis-resolution reasons.

### Setting 20 (transform\_mix): boundary-concentrated curvature

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$$\tilde{g}_{20}(\mathbf{s}) = \text{std}\{\log(1 + s_1)\} + \text{std}\{e^{-s_1/2}\} + \text{std}\{s_1/(1 + s_2)\}. \quad (11)$$

- Every term is monotone in  $s_1$  with curvature maximal on the domain boundary, e.g.  $\partial_{s_1}^2 \log(1 + s_1) = -(1 + s_1)^{-2}$ , largest at  $s_1 = 0$  — exactly where the uniform site draw is sparsest. Hence the slowest  $e(K)$  decay of the quartet (0.21 at  $K = 50$ ).
- Role: the transformations an applied modeller actually writes down (log, exponential decay, ratio); MRTS must spend its resolution where the data density is lowest.

### Setting 21 (nonsmooth): structural non-representability

- With  $(x)_+ = \max(x, 0)$ ,

$$\tilde{g}_{21}(\mathbf{s}) = \text{std}\{(s_1 - 5)_+\} + \text{std}\{|s_2 - 4|\}. \quad (12)$$

- $\tilde{g}_{21} \in C^0 \setminus C^1$ : its gradient jumps across the crease lines  $\{s_1 = 5\}$  and  $\{s_2 = 4\}$ , while every MRTS column is a  $C^\infty$  thin-plate-spline eigenfunction. The projection error therefore *plateaus* instead of vanishing:  $e(30) = 0.074$ ,  $e(40) = 0.064$ ,  $e(50) = 0.063$ .
- Role: the mean-channel counterpart of the dependence settings 3–8 — structure that *no*  $K$  can fully represent, bounding the best MRTS fit away from the oracle at every  $K$ .

### What the spotlight settings show

Settings 16 and 20 are now fitted at  $n = 10$  on the full  $K$  grid; 19 and 21 remain  $n = 3$  pilots. The  $n = 10$  pair carries the weight of what follows.

- **Unthresholded methods gain across the board.** On both  $n = 10$  settings the Gaussian and skew- $t$  methods reduce all three scores: setting 16 to 0.94–0.96 (energy/variogram) and 0.93–0.95 (Brier); setting 20 more strongly, with skew- $t$  K5 the standout (0.70 energy, 0.63 variogram, 0.60 Brier) (Tables 3–5, 7). Under a misspecified error law the mean channel finally moves every score in the same direction — unlike the skew- $t$ -core settings, where these methods were flat.



- **The thresholded methods split by surface — and the  $n = 3$  “inversion” was largely small-sample noise.** At  $n = 10$  the censored methods are *flat* on setting 16 (energy 1.00/1.00, variogram 1.00/1.00 for the K1/K5 variants) — *not* the 1.5–1.7 energy *loss* the  $n = 3$  pilot had reported — yet gain *strongly* on setting 20 (energy 0.53/0.53, variogram 0.42/0.43). The discriminator is where the surface puts its curvature: setting 20 concentrates it where the uniform sites are sparsest, and there the mean basis rescues the censored likelihood; the smoother sigmoid front of setting 16 leaves nothing to rescue. The larger energy-ratio excursions still visible at  $n = 3$  (settings 14, 15, 17, up to 2.9) should be read with that same small- $n$  caution.
- **Brier improves for every method on both  $n = 10$  settings (0.60–0.95);** the realised gains do not follow the  $e(K)$  ordering and stay far above the floor  $R^*$ , both consistent with Finding 3’s sharpness–extrapolation account.

## 6 In progress

1. **Non- $t$  extension analysis (settings 13–21):** the extension exists because all earlier settings share the skew- $t$  core, so at large  $K$  the fitted skew- $t$  is *correctly specified* — a confound for every “MRTS helps/hurts” conclusion. Its nine settings are now fitted and scored ( $n = 10$  on the full  $K$  grid for settings 16 and 20;  $n = 3$  pilots for the other seven; Tables 3–5 and 7; four settings introduced in Section 5), but the systematic write-up — in particular why the thresholded methods, the carriers of every skew- $t$ -core gain, *lose* under some misspecified-error surfaces (energy ratios up to 2.9 at  $n = 3$ , settings 14, 15, 17) yet are flat on setting 16 and gain sharply on setting 20 at  $n = 10$ , while the unthresholded methods post steady gains throughout — is still to be done. Bringing the remaining seven settings to  $n = 10$  is the next step.
2.  **$n$  upgrade:** settings 4–12 and seven of the non- $t$  block are  $n = 3$  pilots (settings 16 and 20 are already at  $n = 10$ ); the headline contrasts (11, 12, and the disambiguation) merit  $n = 10$ . An automated fill script (fit increment  $\rightarrow$  re-score  $\rightarrow$  verify overlap against the pilot cells  $\rightarrow$  delete fits) performs each upgrade; it resumes cleanly after an interruption by re-inventorying the missing  $(m, d, K)$  cells.



Table 2: Anatomy of the DGPs: marginal law of  $Y_t(\mathbf{s})$ , spatial dependence of the error process, and time-averaged true mean  $\bar{\mu}(\mathbf{s})$  (its spatial sd in parentheses; per-dataset values for random means).  $C_{\text{exp}}(h) = 0.9 e^{-h}$  is the shared stationary exponential correlation (Matérn  $\nu = \frac{1}{2}$ ,  $\rho = 1$ , unit diagonal, i.e. a 10% nugget); “skew- $t$ ” is the  $t_3$  core of (1) with  $\lambda = 3$ ;  $*$  denotes convolution with an independent additive field ( $\sigma_{\text{add}} = 3$ );  $f_1, f_2$  are the two cosine bumps. For the non- $t$  block (13–21) the error is  $\sigma_g \varepsilon$  with  $\varepsilon \sim N(0, C_{\text{exp}})$ ,  $\sigma_g = 4$  (so the marginal is Gaussian, sd 4), and every mean surface is orthogonalised against  $[1, s_1, s_2]$  and rescaled to sd = 11.

| ID                                                                                                     | surf_type        | marginal law                                                                                                             | spatial dependence                                                                                                                                    | mean structure $\bar{\mu}(\mathbf{s})$                                                                                               |
|--------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1                                                                                                      | invariant        | skew- $t$                                                                                                                | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + 5f_1 + 3f_2$ (3.7)                                                                                                             |
| 2                                                                                                      | varying          | skew- $t$                                                                                                                | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + w_{1t}f_1 + w_{2t}f_2$ ,<br>( $w_{1t}, w_{2t}$ ) $\sim$<br>$N(0, \text{diag}(25, 9))$ per<br>day; $\bar{\mu} \approx 10$ (0.4) |
| 3                                                                                                      | ns_dependence    | skew- $t * \text{Gaussian}$                                                                                              | $C_{\text{exp}} + \text{low-rank cosine}$<br>$\sum_j \tau_j^2 \phi_j(\mathbf{s}) \phi_j(\mathbf{s}')$ ; non-<br>stationary, negative at long<br>range | $10$ (RE mean-zero) (0.4)                                                                                                            |
| 4                                                                                                      | ps_cov           | skew- $t$ (unchanged)                                                                                                    | Paciorek–Schervish locally<br>anisotropic kernel; scale, as-<br>pect and rotation each vary<br>across the domain                                      | $10$ exactly (0)                                                                                                                     |
| 5                                                                                                      | ps_add           | skew- $t * \text{Gaussian}$                                                                                              | $C_{\text{exp}} + \sigma_{\text{add}}^2 C_{\text{PS}}$ , additive                                                                                     | $10$ (RE mean-zero) (0.4)                                                                                                            |
| 6                                                                                                      | covdep_cov       | skew- $t$ (unchanged)                                                                                                    | isotropic PS kernel with lo-<br>cal range $\rho(\mathbf{s}) = e^{0.7f_1(\mathbf{s})} \in$<br>$[0.5, 2.0]$                                             | $10$ exactly (0)                                                                                                                     |
| 7                                                                                                      | covdep_add       | skew- $t * \text{Gaussian}$                                                                                              | $C_{\text{exp}} + \sigma_{\text{add}}^2 C_{\rho(\mathbf{s})}$ , additive                                                                              | $10$ (RE mean-zero) (0.4)                                                                                                            |
| 8                                                                                                      | fuentes_cov      | skew- $t$ (unchanged)                                                                                                    | Fuentes 4-regime weighted GP<br>mixture, ranges 0.4–4                                                                                                 | $10$ exactly (0)                                                                                                                     |
| 9                                                                                                      | gh_marginal      | skew- $t * \text{std. Tukey } g\text{-}$<br>$\text{and-}h$ ; $g(\mathbf{s}) = 0.8f_1$ ,<br>$h(\mathbf{s}) = 0.12(1+f_2)$ | $C_{\text{exp}}$ in both components<br>(transform perturbs correla-<br>tion only mildly)                                                              | $10$ (standardised mean-<br>zero) (0.4)                                                                                              |
| 10                                                                                                     | lambda_varying   | skew- $t$ with $\lambda(\mathbf{s}) = 3 +$<br>$3f_2 \in [0, 6]$                                                          | $C_{\text{exp}}$ , stationary                                                                                                                         | induced: $10 + \lambda(\mathbf{s}) \bar{z}$ , $\bar{z} =$<br>$\frac{1}{T} \sum_t \sigma_t  z_t $ (2.8)                               |
| 11                                                                                                     | invariant_strong | skew- $t$                                                                                                                | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + 15f_1 + 9f_2$ (11.0)                                                                                                           |
| 12                                                                                                     | mrts_multibump   | skew- $t$                                                                                                                | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + \text{six alternating-sign}$<br>$\text{Gaussian bumps (width}$<br>$1.6)$ (11.0)                                                |
| <i>Non-<math>t</math> extension: Gaussian-GP errors, <math>\sigma_g = 4</math>, mean surface sd 11</i> |                  |                                                                                                                          |                                                                                                                                                       |                                                                                                                                      |
| 13                                                                                                     | franke           | Gaussian                                                                                                                 | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + \text{Franke's function (2}$<br>$\text{large} + 1 \text{ medium bump, 1}$<br>$\text{fine dip)}$ (11.0)                         |
| 14                                                                                                     | ridge_curved     | Gaussian                                                                                                                 | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + \text{curved Gaussian}$<br>$\text{ridge } \exp[-(s'_2 - 0.5 -$<br>$0.25 \sin 2\pi s'_1)^2/2 - 0.12^2]$<br>(11.0)               |
| 15                                                                                                     | annulus          | Gaussian                                                                                                                 | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + \text{Ricker ring } \psi((r -$<br>$3)/1.5), r = \ \mathbf{s} - (5, 5)\ $<br>(11.0)                                             |
| 16                                                                                                     | sigmoid_front    | Gaussian                                                                                                                 | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + \tanh[(\ \mathbf{s} - (2, 2)\  -$<br>$4.5)/1.0]$ arc front (11.0)                                                              |
| 17                                                                                                     | gp_mean          | Gaussian                                                                                                                 | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + \text{GP draw (Matérn}$<br>$\nu=1.5, \text{ range } 2)$ , redrawn<br>per dataset, fixed in $t$ (11.0)                          |
| 18                                                                                                     | franke_sas       | $\sinh\text{-arcsinh}$ ( $\epsilon=0.8$ ,<br>$\delta=1$ ), right-skewed                                                  | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + \text{Franke (identical to}$<br>setting 13) (11.0)                                                                             |
| 19                                                                                                     | poly2            | Gaussian                                                                                                                 | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + \text{2nd-order polynomial}$<br>(bowl + saddle) (11.0)                                                                         |
| 20                                                                                                     | transform_mix    | Gaussian                                                                                                                 | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + \log(1+s_1) + e^{-s_1/2} +$<br>$s_1/(1+s_2)$ (11.0)                                                                            |
| 21                                                                                                     | nonsmooth        | Gaussian                                                                                                                 | $C_{\text{exp}}$ , stationary                                                                                                                         | $10 + \text{hinge}(s_1-5) +  s_2 - 4 $<br>( $C^0$ creases) (11.0)                                                                    |

Table 3: **Energy score**, best over  $K > 0$  relative to the same method’s  $K = 0$  (minimising  $K$  in parentheses). Values  $< 1$ : MRTS helps; bold:  $\bar{R} < 0.75$ .

| ID                                                                                            | non-stationarity                              | $n$ | Gaussian               | Skew- $t$ K1           | Sym- $t$ K1 T                 | Skew- $t$ K5                  | Sym- $t$ K5 T                 |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------|-----|------------------------|------------------------|-------------------------------|-------------------------------|-------------------------------|
| 1                                                                                             | mean (2 bumps, sd 3.7)                        | 10  | 0.994* <sup>(40)</sup> | 0.980* <sup>(40)</sup> | <b>0.579*</b> <sup>(20)</sup> | 0.983* <sup>(50)</sup>        | <b>0.659*</b> <sup>(12)</sup> |
| 3                                                                                             | dependence (low-rank)                         | 10  | 1.000 <sup>(5)</sup>   | 1.000 <sup>(3)</sup>   | 0.993 <sup>(5)</sup>          | 1.000 <sup>(15)</sup>         | 0.911 <sup>(3)</sup>          |
| 4                                                                                             | dependence (PS, repl. $C$ )                   | 3   | 0.999* <sup>(5)</sup>  | 1.000 <sup>(30)</sup>  | 0.889* <sup>(40)</sup>        | 0.996* <sup>(3)</sup>         | 0.846 <sup>(15)</sup>         |
| 5                                                                                             | dependence (PS, additive)                     | 3   | 0.999 <sup>(12)</sup>  | 0.999* <sup>(8)</sup>  | 0.953 <sup>(15)</sup>         | 0.998* <sup>(10)</sup>        | 0.955 <sup>(15)</sup>         |
| 6                                                                                             | dependence ( $\rho(\mathbf{s})$ , repl. $C$ ) | 3   | 1.000 <sup>(10)</sup>  | 0.999 <sup>(10)</sup>  | 0.861* <sup>(20)</sup>        | 0.997* <sup>(12)</sup>        | 0.908 <sup>(50)</sup>         |
| 7                                                                                             | dependence ( $\rho(\mathbf{s})$ , additive)   | 3   | 1.000 <sup>(25)</sup>  | 0.999 <sup>(30)</sup>  | 0.985 <sup>(30)</sup>         | 0.995 <sup>(25)</sup>         | 0.926 <sup>(15)</sup>         |
| 8                                                                                             | dependence (Fuentes, repl. $C$ )              | 3   | 1.000 <sup>(10)</sup>  | 1.000 <sup>(3)</sup>   | 0.848 <sup>(25)</sup>         | 0.998 <sup>(5)</sup>          | <b>0.695</b> <sup>(30)</sup>  |
| 9                                                                                             | tails ( $g$ -and- $h$ )                       | 3   | 0.999* <sup>(5)</sup>  | 1.000 <sup>(10)</sup>  | 0.887 <sup>(5)</sup>          | 0.986 <sup>(25)</sup>         | 0.942 <sup>(5)</sup>          |
| 10                                                                                            | skewness ( $\lambda(\mathbf{s})$ )            | 3   | 0.993* <sup>(50)</sup> | 1.000 <sup>(50)</sup>  | 0.975 <sup>(5)</sup>          | 0.995 <sup>(30)</sup>         | 0.938* <sup>(5)</sup>         |
| 11                                                                                            | mean (2 bumps, sd 11)                         | 3   | 0.976* <sup>(50)</sup> | 0.965* <sup>(50)</sup> | <b>0.199*</b> <sup>(20)</sup> | 0.961* <sup>(50)</sup>        | <b>0.267*</b> <sup>(20)</sup> |
| 12                                                                                            | mean (6 bumps, sd 11)                         | 3   | 0.960* <sup>(50)</sup> | 0.953* <sup>(50)</sup> | <b>0.421*</b> <sup>(40)</sup> | 0.952* <sup>(50)</sup>        | 0.846 <sup>(15)</sup>         |
| <i>Non-<math>t</math> extension (Gaussian-GP errors; skew-<math>t</math> is misspecified)</i> |                                               |     |                        |                        |                               |                               |                               |
| 13                                                                                            | mean, Franke (multiscale)                     | 3   | 0.944* <sup>(50)</sup> | 0.945* <sup>(50)</sup> | 1.198 <sup>(50)</sup>         | 0.942* <sup>(50)</sup>        | 1.179 <sup>(40)</sup>         |
| 14                                                                                            | mean, curved ridge                            | 3   | 0.956* <sup>(40)</sup> | 0.956* <sup>(40)</sup> | 2.263* <sup>(50)</sup>        | 0.953* <sup>(40)</sup>        | 2.938* <sup>(40)</sup>        |
| 15                                                                                            | mean, annulus (Ricker)                        | 3   | 1.028* <sup>(50)</sup> | 1.029* <sup>(50)</sup> | 1.159 <sup>(30)</sup>         | 1.031* <sup>(50)</sup>        | 1.116* <sup>(40)</sup>        |
| 16                                                                                            | mean, sigmoid front                           | 10  | 0.957* <sup>(20)</sup> | 0.957* <sup>(50)</sup> | 0.999 <sup>(3)</sup>          | 0.956* <sup>(50)</sup>        | 1.002 <sup>(3)</sup>          |
| 17                                                                                            | mean, GP draw                                 | 3   | 1.005 <sup>(50)</sup>  | 1.004 <sup>(50)</sup>  | 1.301 <sup>(40)</sup>         | 0.993 <sup>(50)</sup>         | 1.343* <sup>(40)</sup>        |
| 18                                                                                            | mean, Franke + sinh-arcsinh err.              | 3   | 0.941* <sup>(50)</sup> | 0.942* <sup>(50)</sup> | 0.851 <sup>(30)</sup>         | 0.950* <sup>(50)</sup>        | 0.975 <sup>(50)</sup>         |
| 19                                                                                            | mean, poly2 (bowl+saddle)                     | 3   | 0.954* <sup>(30)</sup> | 0.953* <sup>(50)</sup> | 1.343 <sup>(40)</sup>         | 0.946* <sup>(50)</sup>        | 0.973 <sup>(40)</sup>         |
| 20                                                                                            | mean, transform mix                           | 10  | 0.973* <sup>(30)</sup> | 0.971* <sup>(50)</sup> | <b>0.531*</b> <sup>(50)</sup> | <b>0.699*</b> <sup>(50)</sup> | <b>0.527*</b> <sup>(5)</sup>  |
| 21                                                                                            | mean, non-smooth (hinges)                     | 3   | 0.949* <sup>(30)</sup> | 0.947* <sup>(30)</sup> | 1.052 <sup>(30)</sup>         | 0.927* <sup>(30)</sup>        | 1.214 <sup>(30)</sup>         |

Table 4: **Brier score** (mean over the 11 exceedance levels), best over  $K > 0$  relative to own  $K = 0$ . Bold:  $\bar{R} < 0.75$ .

| ID                                                                                            | non-stationarity                              | $n$ | Gaussian               | Skew- $t$ K1           | Sym- $t$ K1 T          | Skew- $t$ K5                  | Sym- $t$ K5 T                 |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------|-----|------------------------|------------------------|------------------------|-------------------------------|-------------------------------|
| 1                                                                                             | mean (2 bumps, sd 3.7)                        | 10  | 1.000 <sup>(3)</sup>   | 0.978* <sup>(50)</sup> | 0.993* <sup>(30)</sup> | 0.984* <sup>(50)</sup>        | 0.971 <sup>(3)</sup>          |
| 3                                                                                             | dependence (low-rank)                         | 10  | 1.000 <sup>(5)</sup>   | 0.999 <sup>(5)</sup>   | 1.000 <sup>(3)</sup>   | 0.996 <sup>(40)</sup>         | 0.993 <sup>(12)</sup>         |
| 4                                                                                             | dependence (PS, repl. $C$ )                   | 3   | 0.998 <sup>(50)</sup>  | 0.998* <sup>(30)</sup> | 0.998* <sup>(30)</sup> | 0.983* <sup>(50)</sup>        | 0.940* <sup>(20)</sup>        |
| 5                                                                                             | dependence (PS, additive)                     | 3   | 1.000 <sup>(3)</sup>   | 0.999 <sup>(12)</sup>  | 1.000 <sup>(8)</sup>   | 0.990 <sup>(15)</sup>         | 0.973 <sup>(5)</sup>          |
| 6                                                                                             | dependence ( $\rho(\mathbf{s})$ , repl. $C$ ) | 3   | 0.998 <sup>(50)</sup>  | 0.999* <sup>(5)</sup>  | 0.997* <sup>(40)</sup> | 0.983* <sup>(12)</sup>        | 0.970 <sup>(10)</sup>         |
| 7                                                                                             | dependence ( $\rho(\mathbf{s})$ , additive)   | 3   | 1.000 <sup>(15)</sup>  | 0.999 <sup>(10)</sup>  | 0.998 <sup>(15)</sup>  | 0.990 <sup>(5)</sup>          | 0.968 <sup>(15)</sup>         |
| 8                                                                                             | dependence (Fuentes, repl. $C$ )              | 3   | 1.000* <sup>(3)</sup>  | 1.000 <sup>(3)</sup>   | 0.999 <sup>(3)</sup>   | 0.997 <sup>(12)</sup>         | 0.962 <sup>(12)</sup>         |
| 9                                                                                             | tails ( $g$ -and- $h$ )                       | 3   | 1.000 <sup>(12)</sup>  | 1.000 <sup>(15)</sup>  | 0.998 <sup>(40)</sup>  | 0.996 <sup>(10)</sup>         | 0.967 <sup>(12)</sup>         |
| 10                                                                                            | skewness ( $\lambda(\mathbf{s})$ )            | 3   | 0.988 <sup>(50)</sup>  | 0.998 <sup>(50)</sup>  | 0.949* <sup>(50)</sup> | 0.988 <sup>(30)</sup>         | 0.972 <sup>(30)</sup>         |
| 11                                                                                            | mean (2 bumps, sd 11)                         | 3   | 1.000 <sup>(3)</sup>   | 1.000 <sup>(3)</sup>   | 0.952* <sup>(30)</sup> | 1.004 <sup>(3)</sup>          | 0.997 <sup>(3)</sup>          |
| 12                                                                                            | mean (6 bumps, sd 11)                         | 3   | 0.930 <sup>(50)</sup>  | 0.925* <sup>(50)</sup> | 0.880* <sup>(50)</sup> | 0.913* <sup>(50)</sup>        | 0.884* <sup>(10)</sup>        |
| <i>Non-<math>t</math> extension (Gaussian-GP errors; skew-<math>t</math> is misspecified)</i> |                                               |     |                        |                        |                        |                               |                               |
| 13                                                                                            | mean, Franke (multiscale)                     | 3   | 0.974* <sup>(50)</sup> | 0.972* <sup>(50)</sup> | 0.956 <sup>(50)</sup>  | 0.950* <sup>(50)</sup>        | 0.936* <sup>(30)</sup>        |
| 14                                                                                            | mean, curved ridge                            | 3   | 0.990 <sup>(50)</sup>  | 0.989* <sup>(50)</sup> | 0.979 <sup>(40)</sup>  | 0.895* <sup>(50)</sup>        | 0.964* <sup>(40)</sup>        |
| 15                                                                                            | mean, annulus (Ricker)                        | 3   | 1.143* <sup>(50)</sup> | 1.142* <sup>(50)</sup> | 0.874* <sup>(40)</sup> | 1.144* <sup>(50)</sup>        | 0.924* <sup>(40)</sup>        |
| 16                                                                                            | mean, sigmoid front                           | 10  | 0.950* <sup>(25)</sup> | 0.949* <sup>(25)</sup> | 0.916* <sup>(25)</sup> | 0.935* <sup>(50)</sup>        | 0.912* <sup>(30)</sup>        |
| 17                                                                                            | mean, GP draw                                 | 3   | 0.984 <sup>(40)</sup>  | 0.982 <sup>(40)</sup>  | 0.913* <sup>(50)</sup> | 0.971 <sup>(40)</sup>         | 0.867* <sup>(50)</sup>        |
| 18                                                                                            | mean, Franke + sinh-arcsinh err.              | 3   | 0.970* <sup>(50)</sup> | 0.971* <sup>(50)</sup> | 0.920* <sup>(30)</sup> | 0.960* <sup>(50)</sup>        | 0.887* <sup>(30)</sup>        |
| 19                                                                                            | mean, poly2 (bowl+saddle)                     | 3   | 0.931* <sup>(50)</sup> | 0.927* <sup>(50)</sup> | 0.907* <sup>(40)</sup> | 0.896* <sup>(50)</sup>        | 0.885* <sup>(40)</sup>        |
| 20                                                                                            | mean, transform mix                           | 10  | 0.930* <sup>(30)</sup> | 0.927* <sup>(30)</sup> | 0.786* <sup>(15)</sup> | <b>0.601*</b> <sup>(40)</sup> | <b>0.663*</b> <sup>(15)</sup> |
| 21                                                                                            | mean, non-smooth (hinges)                     | 3   | 0.918* <sup>(30)</sup> | 0.917* <sup>(30)</sup> | 0.896* <sup>(30)</sup> | 0.875* <sup>(30)</sup>        | 0.852* <sup>(40)</sup>        |

Table 5: **Variogram score** ( $p = 1/2$ ), best over  $K > 0$  relative to own  $K = 0$ . Bold:  $\bar{R} < 0.75$ .

| ID                                                                                            | non-stationarity                              | $n$ | Gaussian               | Skew- $t$ K1           | Sym- $t$ K1 T                 | Skew- $t$ K5                  | Sym- $t$ K5 T                 |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------|-----|------------------------|------------------------|-------------------------------|-------------------------------|-------------------------------|
| 1                                                                                             | mean (2 bumps, sd 3.7)                        | 10  | 0.994* <sub>(40)</sub> | 0.975* <sub>(40)</sub> | <b>0.617*</b> <sub>(15)</sub> | 0.981* <sub>(50)</sub>        | <b>0.712</b> <sub>(12)</sub>  |
| 3                                                                                             | dependence (low-rank)                         | 10  | 1.000 <sub>(5)</sub>   | 1.000 <sub>(3)</sub>   | 0.984 <sub>(5)</sub>          | 0.999 <sub>(25)</sub>         | 0.894 <sub>(3)</sub>          |
| 4                                                                                             | dependence (PS, repl. $C$ )                   | 3   | 0.999* <sub>(10)</sub> | 1.000 <sub>(15)</sub>  | 0.988 <sub>(15)</sub>         | 0.997 <sub>(10)</sub>         | 0.990 <sub>(3)</sub>          |
| 5                                                                                             | dependence (PS, additive)                     | 3   | 0.999* <sub>(12)</sub> | 0.999* <sub>(8)</sub>  | 0.963 <sub>(50)</sub>         | 0.998 <sub>(15)</sub>         | 0.986 <sub>(15)</sub>         |
| 6                                                                                             | dependence ( $\rho(\mathbf{s})$ , repl. $C$ ) | 3   | 0.999 <sub>(10)</sub>  | 0.999 <sub>(50)</sub>  | 0.920* <sub>(40)</sub>        | 0.997 <sub>(12)</sub>         | 0.984 <sub>(10)</sub>         |
| 7                                                                                             | dependence ( $\rho(\mathbf{s})$ , additive)   | 3   | 1.000 <sub>(10)</sub>  | 1.000 <sub>(30)</sub>  | 0.995 <sub>(3)</sub>          | 0.995 <sub>(5)</sub>          | 0.960 <sub>(15)</sub>         |
| 8                                                                                             | dependence (Fuentes, repl. $C$ )              | 3   | 1.000 <sub>(8)</sub>   | 1.000 <sub>(3)</sub>   | 0.887 <sub>(40)</sub>         | 1.000 <sub>(3)</sub>          | 0.847 <sub>(5)</sub>          |
| 9                                                                                             | tails ( $g$ -and- $h$ )                       | 3   | 1.000 <sub>(3)</sub>   | 1.000 <sub>(5)</sub>   | 0.865 <sub>(5)</sub>          | 0.982 <sub>(25)</sub>         | 0.947 <sub>(5)</sub>          |
| 10                                                                                            | skewness ( $\lambda(\mathbf{s})$ )            | 3   | 0.990 <sub>(40)</sub>  | 0.999 <sub>(30)</sub>  | 1.000 <sub>(3)</sub>          | 0.993 <sub>(30)</sub>         | 0.905* <sub>(10)</sub>        |
| 11                                                                                            | mean (2 bumps, sd 11)                         | 3   | 0.963* <sub>(50)</sub> | 0.943* <sub>(50)</sub> | <b>0.201*</b> <sub>(10)</sub> | 0.935* <sub>(50)</sub>        | <b>0.244*</b> <sub>(20)</sub> |
| 12                                                                                            | mean (6 bumps, sd 11)                         | 3   | 0.938* <sub>(50)</sub> | 0.911* <sub>(50)</sub> | <b>0.354*</b> <sub>(40)</sub> | 0.915* <sub>(50)</sub>        | 0.909 <sub>(15)</sub>         |
| <i>Non-<math>t</math> extension (Gaussian-GP errors; skew-<math>t</math> is misspecified)</i> |                                               |     |                        |                        |                               |                               |                               |
| 13                                                                                            | mean, Franke (multiscale)                     | 3   | 0.912* <sub>(50)</sub> | 0.913* <sub>(50)</sub> | 1.164* <sub>(30)</sub>        | 0.912* <sub>(50)</sub>        | 1.085 <sub>(30)</sub>         |
| 14                                                                                            | mean, curved ridge                            | 3   | 0.891* <sub>(40)</sub> | 0.891* <sub>(40)</sub> | 2.092* <sub>(50)</sub>        | 0.913* <sub>(40)</sub>        | 2.571* <sub>(40)</sub>        |
| 15                                                                                            | mean, annulus (Ricker)                        | 3   | 1.032* <sub>(50)</sub> | 1.031* <sub>(50)</sub> | 1.448* <sub>(30)</sub>        | 1.035* <sub>(50)</sub>        | 1.444* <sub>(40)</sub>        |
| 16                                                                                            | mean, sigmoid front                           | 10  | 0.941* <sub>(20)</sub> | 0.941* <sub>(50)</sub> | 0.998 <sub>(3)</sub>          | 0.938* <sub>(20)</sub>        | 1.000 <sub>(3)</sub>          |
| 17                                                                                            | mean, GP draw                                 | 3   | 1.000 <sub>(50)</sub>  | 0.998 <sub>(50)</sub>  | 1.205 <sub>(40)</sub>         | 0.983 <sub>(50)</sub>         | 1.218 <sub>(40)</sub>         |
| 18                                                                                            | mean, Franke + sinh-arcsinh err.              | 3   | 0.905* <sub>(50)</sub> | 0.906* <sub>(50)</sub> | <b>0.726*</b> <sub>(30)</sub> | 0.919* <sub>(50)</sub>        | 0.854 <sub>(30)</sub>         |
| 19                                                                                            | mean, poly2 (bowl+saddle)                     | 3   | 0.935* <sub>(30)</sub> | 0.935* <sub>(50)</sub> | 1.542 <sub>(40)</sub>         | 0.928* <sub>(50)</sub>        | 1.020 <sub>(40)</sub>         |
| 20                                                                                            | mean, transform mix                           | 10  | 0.963* <sub>(30)</sub> | 0.961* <sub>(50)</sub> | <b>0.420*</b> <sub>(15)</sub> | <b>0.628*</b> <sub>(50)</sub> | <b>0.426*</b> <sub>(5)</sub>  |
| 21                                                                                            | mean, non-smooth (hinges)                     | 3   | 0.934* <sub>(30)</sub> | 0.932* <sub>(30)</sub> | 1.055 <sub>(30)</sub>         | 0.904* <sub>(30)</sub>        | 1.209 <sub>(30)</sub>         |

Table 6: **Relative Brier score at every  $K_{\text{MRTS}}$ , K1-knot methods (Gaussian, Skew- $t$  K1, Sym- $t$  K1 T), skew- $t$  settings (vs the same method’s own  $K = 0$ , mean over datasets; \* marks  $|\bar{R} - 1| > 2\widehat{\text{SE}}$ ). A dash means that  $K$  was not fitted for that setting (settings 13–21 use only  $K \in \{30, 40, 50\}$ ). Isolated large values of Sym- $t$  K5 T are the exponential-fallback outliers discussed in the Caveats and are reported as-is.**

| ID                             | $n$ | $K=3$  | $K=5$  | $K=8$  | $K=10$ | $K=12$ | $K=15$ | $K=20$ | $K=25$ | $K=30$ | $K=40$ | $K=50$ |
|--------------------------------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| <i>Gaussian</i>                |     |        |        |        |        |        |        |        |        |        |        |        |
| 1                              | 10  | 1.000  | 1.016* | 1.023* | 1.022* | 1.019* | 1.016* | 1.009* | 1.008* | 1.005  | 1.003  | 1.003  |
| 3                              | 10  | 1.000  | 1.000  | 1.001* | 1.001* | 1.000  | 1.000  | 1.000  | 1.001  | 1.001  | 1.001* | 1.002* |
| 4                              | 3   | 1.000  | 0.999* | 0.999* | 1.000  | 1.000  | 1.000  | 0.999  | 0.999  | 0.999  | 0.999  | 0.998  |
| 5                              | 3   | 1.000  | 1.001  | 1.001  | 1.000  | 1.001* | 1.000  | 1.001* | 1.002* | 1.002* | 1.001* | 1.004* |
| 6                              | 3   | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 0.999  | 1.000  | 0.999  | 0.998* | 0.998  |
| 7                              | 3   | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.002* | 1.001  | 1.000  | 1.000  | 1.000  |
| 8                              | 3   | 1.000* | 1.001  | 1.001  | 1.001  | 1.000  | 1.000  | 1.002  | 1.002  | 1.002  | 1.003  | 1.004* |
| 9                              | 3   | 1.000* | 1.000  | 1.000  | 1.000* | 1.000  | 1.000  | 1.000  | 1.001  | 1.001  | 1.002  | 1.001  |
| 10                             | 3   | 1.000  | 1.003  | 1.008  | 1.004  | 1.000  | 0.997  | 0.995  | 0.995  | 0.989  | 0.988  | 0.988  |
| 11                             | 3   | 1.000  | 1.059* | 1.078* | 1.101* | 1.114* | 1.095* | 1.069* | 1.051* | 1.022  | 1.001  | 1.000  |
| 12                             | 3   | 1.000  | 1.003* | 1.000  | 0.993  | 0.980  | 1.019  | 1.049  | 1.045  | 0.988  | 0.947  | 0.930  |
| <i>Skew-<math>t</math> K1</i>  |     |        |        |        |        |        |        |        |        |        |        |        |
| 1                              | 10  | 1.000  | 0.996  | 1.006  | 1.006  | 1.001  | 0.996  | 0.987* | 0.985* | 0.981* | 0.978* | 0.978* |
| 3                              | 10  | 1.000  | 0.999  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  |
| 4                              | 3   | 1.000* | 1.000  | 0.999  | 1.000  | 1.000  | 1.000  | 0.999  | 0.999  | 0.998* | 0.999  | 0.999  |
| 5                              | 3   | 1.000  | 1.000  | 0.999  | 1.000  | 0.999  | 1.000  | 1.000  | 1.001  | 1.001  | 1.001  | 1.002  |
| 6                              | 3   | 1.000  | 0.999* | 0.999* | 1.000  | 1.001* | 0.999  | 1.000  | 1.000  | 1.000  | 0.999  | 1.000  |
| 7                              | 3   | 1.000  | 1.000  | 1.001  | 0.999  | 1.000  | 1.000  | 1.001  | 1.000  | 1.000  | 1.001  | 1.001  |
| 8                              | 3   | 1.000  | 1.000  | 1.001  | 1.000  | 1.001  | 1.001* | 1.001  | 1.000  | 1.001  | 1.001* | 1.002* |
| 9                              | 3   | 1.001  | 1.001  | 1.001* | 1.001  | 1.001  | 1.000  | 1.001  | 1.001  | 1.001  | 1.001  | 1.001  |
| 10                             | 3   | 0.999  | 1.003  | 1.008* | 1.008  | 1.005  | 1.003  | 1.002  | 1.001  | 0.999  | 0.998  | 0.998  |
| 11                             | 3   | 1.000  | 1.052* | 1.074* | 1.103* | 1.163* | 1.138* | 1.110* | 1.086* | 1.048  | 1.017  | 1.014  |
| 12                             | 3   | 1.000  | 1.005* | 0.999  | 0.992  | 0.965  | 1.022  | 1.024  | 1.020  | 0.980  | 0.946  | 0.925* |
| <i>Sym-<math>t</math> K1 T</i> |     |        |        |        |        |        |        |        |        |        |        |        |
| 1                              | 10  | 1.000  | 1.003  | 1.006  | 1.008* | 1.003  | 1.001  | 0.996  | 0.994* | 0.993* | 0.993* | 0.994* |
| 3                              | 10  | 1.000  | 1.000  | 1.001  | 1.001  | 1.001  | 1.001  | 1.001  | 1.001  | 1.002  | 1.002  | 1.004* |
| 4                              | 3   | 1.000  | 0.998* | 0.999  | 0.999  | 1.000  | 0.999  | 1.000  | 0.999  | 0.998* | 1.000  | 0.999  |
| 5                              | 3   | 1.000* | 1.000  | 1.000  | 1.000  | 1.000  | 1.001  | 1.000  | 1.001  | 1.002* | 1.004  | 1.007  |
| 6                              | 3   | 1.000  | 1.002* | 1.000  | 1.000  | 1.001  | 0.999* | 0.999  | 1.000  | 1.001  | 0.997* | 1.000  |
| 7                              | 3   | 1.000* | 1.000  | 1.000  | 1.000  | 1.000  | 0.998  | 1.001  | 1.000  | 1.000  | 0.999  | 1.001  |
| 8                              | 3   | 0.999  | 0.999  | 0.999  | 1.002  | 1.000  | 1.001  | 1.003  | 1.008  | 1.009  | 1.012  | 1.010* |
| 9                              | 3   | 1.000  | 1.001  | 1.001  | 1.000  | 1.001  | 1.001* | 1.000  | 1.001* | 1.002  | 0.998  | 1.000  |
| 10                             | 3   | 0.999  | 0.993* | 0.980* | 0.985  | 0.975  | 0.970  | 0.963* | 0.957* | 0.949* | 0.954* | 0.949* |
| 11                             | 3   | 0.999  | 1.016  | 1.021  | 1.039* | 1.008  | 0.982  | 0.977* | 0.959* | 0.952* | 0.954* | 0.959* |
| 12                             | 3   | 1.000  | 1.006* | 0.997  | 0.907* | 0.893* | 0.899* | 0.896* | 0.918* | 0.892* | 0.896* | 0.880* |

Table 7: **Relative Brier score at every  $K_{\text{MRTS}}$** , K1-knot methods (Gaussian, Skew- $t$  K1, Sym- $t$  K1 T), non- $t$  extension (Gaussian-GP errors) (vs the same method’s own  $K = 0$ , mean over datasets; \* marks  $|\bar{R} - 1| > 2\widehat{\text{SE}}$ ). A dash means that  $K$  was not fitted for that setting (settings 13–21 use only  $K \in \{30, 40, 50\}$ ). Isolated large values of Sym- $t$  K5 T are the exponential-fallback outliers discussed in the Caveats and are reported as-is.

| ID                             | $n$ | $K=3$  | $K=5$  | $K=8$  | $K=10$ | $K=12$ | $K=15$ | $K=20$ | $K=25$ | $K=30$ | $K=40$ | $K=50$ |
|--------------------------------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| <i>Gaussian</i>                |     |        |        |        |        |        |        |        |        |        |        |        |
| 13                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.025  | 0.999  | 0.974* |
| 14                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.052* | 1.059* | 0.990  |
| 15                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.182* | 1.172* | 1.143* |
| 16                             | 10  | 1.000  | 0.991* | 0.992  | 0.996  | 1.009  | 0.986* | 0.960* | 0.950* | 0.955* | 0.952* | 0.951* |
| 17                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.008  | 0.984  | 0.991  |
| 18                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.020  | 0.997  | 0.970* |
| 19                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.935* | 0.935* | 0.931* |
| 20                             | 10  | 0.998* | 1.040* | 1.147* | 1.148* | 1.101* | 1.008  | 1.044* | 1.003  | 0.930* | 0.954* | 0.955* |
| 21                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.918* | 0.921* | 0.923* |
| <i>Skew-<math>t</math> K1</i>  |     |        |        |        |        |        |        |        |        |        |        |        |
| 13                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.024  | 0.998  | 0.972* |
| 14                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.055* | 1.061* | 0.989* |
| 15                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.180* | 1.169* | 1.142* |
| 16                             | 10  | 1.000  | 0.990* | 0.992  | 0.997  | 1.010  | 0.986* | 0.958* | 0.949* | 0.954* | 0.951* | 0.951* |
| 17                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.006  | 0.982  | 0.988  |
| 18                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.022  | 0.999  | 0.971* |
| 19                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.932* | 0.932* | 0.927* |
| 20                             | 10  | 0.998* | 1.041* | 1.149* | 1.150* | 1.097* | 1.004  | 1.041* | 1.000  | 0.927* | 0.951* | 0.951* |
| 21                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.917* | 0.917* | 0.920* |
| <i>Sym-<math>t</math> K1 T</i> |     |        |        |        |        |        |        |        |        |        |        |        |
| 13                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.960  | 0.989  | 0.956  |
| 14                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.079* | 0.979  | 0.980  |
| 15                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.945* | 0.874* | 0.952  |
| 16                             | 10  | 0.999  | 0.995  | 0.973* | 0.975* | 0.932* | 0.937* | 0.921* | 0.916* | 0.916* | 0.924* | 0.925* |
| 17                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.930  | 0.917* | 0.913* |
| 18                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.920* | 0.931* | 0.946  |
| 19                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.912* | 0.907* | 0.930* |
| 20                             | 10  | 0.996* | 0.949* | 0.976* | 0.972* | 1.123* | 0.786* | 0.802* | 0.828* | 0.815* | 0.874* | 0.985  |
| 21                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.896* | 0.907* | 0.909* |

Table 8: **Relative Brier score at every  $K_{\text{MRTS}}$ , K5-knot methods (Skew- $t$  K5, Sym- $t$  K5 T), all settings (vs the same method’s own  $K = 0$ , mean over datasets; \* marks  $|\bar{R} - 1| > 2\widehat{\text{SE}}$ ). A dash means that  $K$  was not fitted for that setting (settings 13–21 use only  $K \in \{30, 40, 50\}$ ). Isolated large values of Sym- $t$  K5 T are the exponential-fallback outliers discussed in the Caveats and are reported as-is.**

| ID                             | $n$ | $K=3$  | $K=5$  | $K=8$  | $K=10$ | $K=12$ | $K=15$ | $K=20$ | $K=25$ | $K=30$ | $K=40$ | $K=50$ |
|--------------------------------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| <i>Skew-<math>t</math> K5</i>  |     |        |        |        |        |        |        |        |        |        |        |        |
| 1                              | 10  | 0.995  | 1.004  | 1.015* | 1.008  | 1.010  | 1.011* | 0.993  | 0.994  | 0.991  | 0.992  | 0.984* |
| 3                              | 10  | 0.999  | 1.004  | 1.006  | 0.998  | 1.001  | 0.997  | 1.000  | 0.998  | 1.003  | 0.996  | 1.000  |
| 4                              | 3   | 0.989* | 0.993  | 0.988  | 0.994  | 0.992  | 0.994  | 0.987  | 0.985* | 0.994  | 0.988  | 0.983* |
| 5                              | 3   | 0.996  | 0.994  | 0.992* | 0.992  | 0.994  | 0.990  | 0.993  | 0.997  | 0.996  | 0.993  | 0.997  |
| 6                              | 3   | 0.998  | 1.001  | 1.000  | 0.996  | 0.983* | 0.996  | 0.986* | 0.989  | 1.002  | 0.988  | 1.005  |
| 7                              | 3   | 0.993  | 0.990  | 0.997  | 0.994  | 0.994  | 0.994  | 0.994  | 0.993  | 0.995  | 0.995  | 1.002  |
| 8                              | 3   | 0.999  | 0.999  | 1.007  | 1.011* | 0.997  | 1.002  | 1.015* | 1.007* | 1.005  | 1.012* | 1.008  |
| 9                              | 3   | 1.005  | 1.007  | 0.999  | 0.996  | 1.007  | 1.001  | 1.003  | 1.003  | 1.001  | 1.005  | 1.003  |
| 10                             | 3   | 0.999  | 1.002* | 1.006* | 1.008  | 1.004  | 1.005  | 1.002  | 0.994  | 0.988  | 0.998  | 0.996  |
| 11                             | 3   | 1.004  | 1.088* | 1.096* | 1.123* | 1.184* | 1.155* | 1.142* | 1.112  | 1.073  | 1.038  | 1.036  |
| 12                             | 3   | 0.992  | 1.034  | 0.995  | 1.004  | 0.962  | 0.976  | 1.017  | 1.022  | 0.965  | 0.927* | 0.913* |
| 13                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.001  | 0.977* | 0.950* |
| 14                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.983* | 0.981* | 0.895* |
| 15                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.177* | 1.168* | 1.144* |
| 16                             | 10  | 0.995* | 0.977* | 0.975* | 0.985* | 0.996  | 0.971* | 0.945* | 0.935* | 0.941* | 0.936* | 0.935* |
| 17                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.996  | 0.971  | 0.980  |
| 18                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.018* | 0.992  | 0.960* |
| 19                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.900* | 0.899* | 0.896* |
| 20                             | 10  | 1.000  | 0.971* | 0.997  | 0.959* | 0.837* | 0.727* | 0.719* | 0.663* | 0.617* | 0.601* | 0.604* |
| 21                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.875* | 0.877* | 0.879* |
| <i>Sym-<math>t</math> K5 T</i> |     |        |        |        |        |        |        |        |        |        |        |        |
| 1                              | 10  | 0.971  | 1.057  | 1.073  | 1.097* | 1.055  | 1.023  | 1.050  | 1.085  | 1.131  | 1.073  | 1.165* |
| 3                              | 10  | 0.997  | 1.011  | 1.001  | 1.009  | 0.993  | 1.013  | 1.068  | 1.066  | 1.007  | 1.124  | 1.160  |
| 4                              | 3   | 0.981* | 0.980  | 0.999  | 0.998  | 1.022  | 0.994  | 0.940* | 0.995  | 1.056  | 0.993  | 0.961  |
| 5                              | 3   | 1.021  | 0.973  | 0.975  | 0.974  | 0.987  | 0.980  | 0.977  | 0.973  | 1.023  | 1.009  | 1.026  |
| 6                              | 3   | 0.980  | 0.978  | 0.994  | 0.970  | 0.988  | 1.020  | 0.997  | 1.028  | 1.033* | 0.998  | 1.045* |
| 7                              | 3   | 0.981  | 0.990  | 0.987  | 0.972  | 0.993  | 0.968  | 1.011  | 0.993  | 1.029  | 1.017  | 1.021  |
| 8                              | 3   | 0.967  | 1.005  | 0.990  | 0.999  | 0.962  | 1.019  | 1.063* | 1.050  | 1.084  | 0.988  | 1.063  |
| 9                              | 3   | 0.992  | 0.982  | 1.015  | 0.991  | 0.967  | 1.008  | 0.998  | 1.040* | 1.029  | 1.045  | 0.974  |
| 10                             | 3   | 0.990  | 1.005  | 0.996  | 1.027  | 1.004  | 0.994  | 0.988  | 1.010  | 0.972  | 0.977  | 0.972  |
| 11                             | 3   | 0.997  | 1.078  | 1.182* | 1.114* | 1.163* | 1.120* | 1.104  | 1.133  | 1.030  | 1.086  | 1.063  |
| 12                             | 3   | 0.994* | 0.998  | 0.999  | 0.884* | 0.903* | 0.966  | 0.917* | 1.321  | 1.012  | 1.061  | 1.587  |
| 13                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.936* | 0.947  | 0.956  |
| 14                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 1.105* | 0.964* | 1.027  |
| 15                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.924* | 0.924* | 0.935* |
| 16                             | 10  | 1.000  | 0.992  | 0.967* | 0.971* | 0.922* | 0.927* | 0.918* | 0.913* | 0.912* | 0.919* | 0.949* |
| 17                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.907* | 0.903* | 0.867* |
| 18                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.887* | 0.962  | 0.893* |
| 19                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.904* | 0.885* | 1.067  |
| 20                             | 10  | 1.002  | 0.910* | 0.817* | 0.792* | 0.920* | 0.663* | 0.688* | 0.701* | 0.693* | 0.781* | 0.690* |
| 21                             | 3   | –      | –      | –      | –      | –      | –      | –      | –      | 0.853* | 0.852* | 0.854* |

Table 9: Oracle Brier floor  $R^*$  per setting (`oracle_brier_ceiling.R`; Monte Carlo, 3 datasets).

| setting             | $sd(\bar{\mu})$ | $R^*$        | verdict             |
|---------------------|-----------------|--------------|---------------------|
| 1 invariant         | 3.68            | 0.856        | weak headroom (14%) |
| 2 varying           | 0.42            | 1.000        | dead end            |
| 3 ns_dependence     | 0.41            | 0.999        | dead end            |
| 4 ps_cov            | 0.00            | 1.000        | dead end            |
| 5 ps_add            | 0.39            | 0.999        | dead end            |
| 6 covdep_cov        | 0.00            | 1.000        | dead end            |
| 7 covdep_add        | 0.41            | 1.000        | dead end            |
| 8 fuentes_cov       | 0.00            | 1.000        | dead end            |
| 9 gh_marginal       | 0.43            | 1.000        | dead end            |
| 10 lambda_varying   | 2.80            | <b>1.120</b> | <b>harmful</b>      |
| 11 invariant_strong | 11.04           | 0.439        | positive control    |
| 12 mrts_multibump   | 11.00           | 0.495        | positive control    |

Table 10: Setting 11, relative score vs  $K_{\text{MRTS}}$  ( $n = 3$ ): the thresholded symmetric- $t$  methods against the unthresholded skew- $t$ .

| score  | method                 | $K=3$ | $K=5$        | $K=8$ | $K=10$ | $K=12$       | $K=15$ | $K=20$       | $K=25$ | $K=30$ | $K=40$ | $K=50$ |
|--------|------------------------|-------|--------------|-------|--------|--------------|--------|--------------|--------|--------|--------|--------|
| energy | Skew- $t$ K1 (no thr.) | 0.999 | 1.013        | 1.022 | 1.027  | 1.023        | 1.017  | 0.996        | 0.984  | 0.978  | 0.966  | 0.965  |
|        | Sym- $t$ K1 (thr.)     | 0.990 | <b>0.263</b> | 0.266 | 0.229  | 0.201        | 0.244  | <b>0.199</b> | 0.256  | 0.233  | 0.268  | 0.305  |
|        | Sym- $t$ K5 (thr.)     | 1.001 | 0.317        | 0.539 | 0.374  | 0.342        | 0.462  | <b>0.267</b> | 0.333  | 0.359  | 0.906  | 0.860  |
| Brier  | Skew- $t$ K1 (no thr.) | 1.000 | 1.052        | 1.074 | 1.103  | <b>1.163</b> | 1.138  | 1.110        | 1.086  | 1.048  | 1.017  | 1.014  |
|        | Sym- $t$ K1 (thr.)     | 0.999 | 1.016        | 1.021 | 1.039  | 1.008        | 0.982  | 0.977        | 0.959  | 0.952  | 0.954  | 0.959  |
|        | Sym- $t$ K5 (thr.)     | 0.997 | 1.078        | 1.182 | 1.114  | 1.163        | 1.120  | 1.104        | 1.133  | 1.030  | 1.086  | 1.063  |

Table 11: Disambiguation on setting 1 ( $n = 10$ ): relative energy score vs  $K_{\text{MRTS}}$ , and the *absolute* energy score at  $K = 0$ . Method pairs (3, 11) and (5, 12) differ *only* in the POT threshold.

|    | method                      | $K=5$ | $K=10$ | $K=15$ | $K=20$       | $K=25$       | abs. ES at $K=0$ |
|----|-----------------------------|-------|--------|--------|--------------|--------------|------------------|
| 3  | Sym- $t$ K1, thr. $q(0.80)$ | 0.630 | 0.617  | 0.589  | <b>0.579</b> | 0.596        | 21.8             |
| 11 | Sym- $t$ K1, no threshold   | 0.996 | 0.997  | 0.991  | 0.990        | 0.989        | 8.5              |
| 5  | Sym- $t$ K5, thr. $q(0.80)$ | 0.688 | 1.113  | 0.773  | 0.718        | <b>0.697</b> | 137.1            |
| 12 | Sym- $t$ K5, no threshold   | 0.994 | 1.008  | 0.993  | 0.997        | 0.997        | 9.3              |
| 2  | Skew- $t$ K1 (reference)    | 0.994 | 0.994  | 0.985  | 0.983        | 0.982        | 8.5              |

Table 12: The spotlight quartet: noise-free projection error  $e(K)$  of (8) and oracle Brier floor  $R^*$  (Monte Carlo, Gaussian error,  $\sigma_g = 4$ ; `nont_precheck.R`).

| ID | surf_type     | $e(5)$ | $e(15)$ | $e(30)$ | $e(50)$ | $K \rightarrow \infty$ behaviour | $R^*$ |
|----|---------------|--------|---------|---------|---------|----------------------------------|-------|
| 16 | sigmoid_front | 0.591  | 0.250   | 0.131   | 0.094   | ↓, elbow at $K \approx 40$ –50   | 0.559 |
| 19 | poly2         | 0.226  | 0.054   | 0.030   | 0.015   | ↓, saturated by $K \approx 12$   | 0.286 |
| 20 | transform_mix | 0.895  | 0.506   | 0.272   | 0.214   | ↓, slowest decay                 | 0.396 |
| 21 | nonsmooth     | 0.469  | 0.159   | 0.074   | 0.063   | plateau $\approx 0.063 > 0$      | 0.431 |